插图英语阅读小册

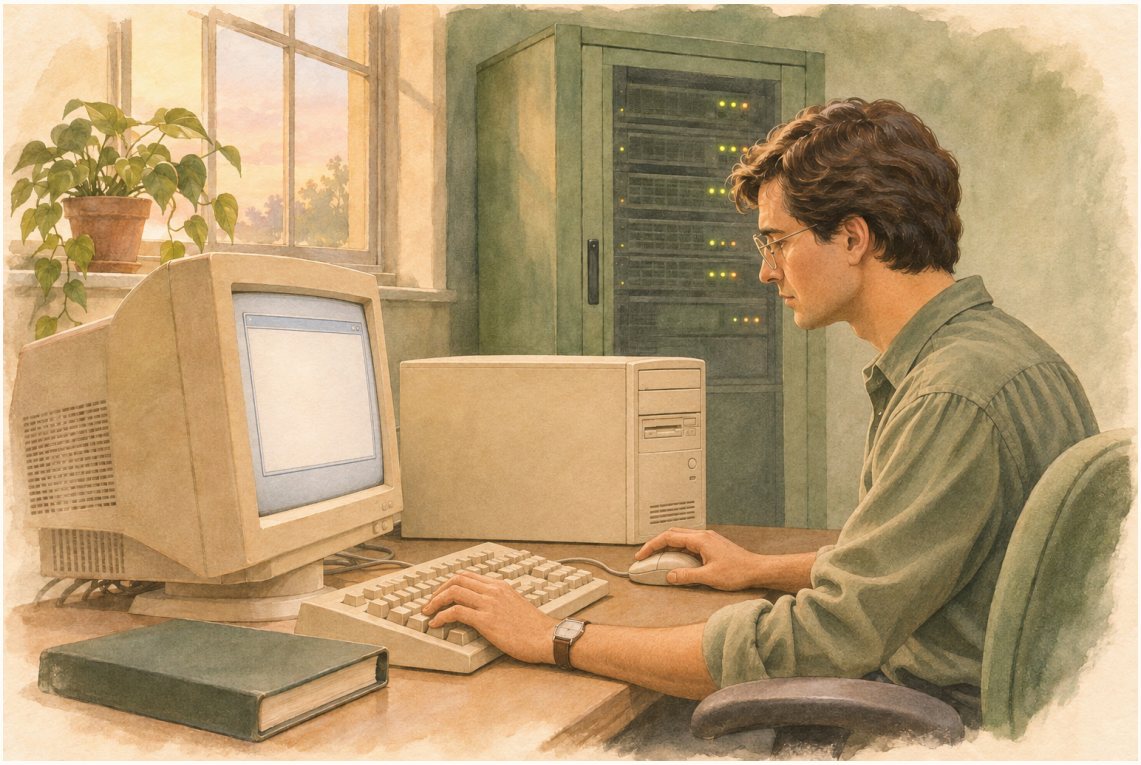
第2册 · 十篇故事

按推荐顺序阅读，保留原篇号。点击篇目即可跳转；每篇依次包含插图、英文原文、中文译文、词汇与回想。

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千年虫：被成功抹去的危机

Y2K: The Crisis That Success Erased



危机没有发生，能说明预防多余吗？

英文原文

The Y2K bug, or millennium bug, began as a shortcut. When memory and hardware were costly, programmers stored years as two digits, writing "99" for 1999. That compact code hid an ambiguity: a machine might read "00" as 1900, corrupting calculations of interest, ages, or deadlines.

Governments and firms braced for chaos. In February 1998 President Clinton created a council to coordinate America's effort. Consultants audited legacy systems and patched faulty date logic. In November 1999 the Commerce Department estimated American governments and enterprises would spend about $100bn; projections of the global bill ran to hundreds of billions.

Then midnight struck on 31 December 1999, and little went wrong. Glitches that did crop up were sporadic and mostly minor. Radiation-monitoring equipment faltered at Japan's Shika nuclear plant without any threat to safety, and a Pentagon ground system that processes spy-satellite data failed. In February 2000 a Senate committee concluded that the expenditure had been justified.

Many saw it differently: the threat had been exaggerated, even a hoax. Yet calm prevailed largely because so much had been fixed. This is the preparedness paradox: when precautions work, the danger they ward off never appears, so it looks imaginary. Tedious repairs earned mockery rather than gratitude.

Critics noted that Italy and South Korea spent far less yet fared no worse. The comparison misleads: those low figures often omitted what firms spent, and much of the remedial work replaced obsolete systems overdue for upgrading. Nobody can say which dollars were essential and which excessive. Prevention that works may leave nothing but an invoice, to be weighed against a catastrophe that might have been.

中文译文

千年虫漏洞最初源于一种省事的做法。当内存和硬件还很昂贵时，程序员只用两位数字储存年份，例如以“99”代表1999年。这种精简代码暗藏歧义：计算机可能把“00”理解成1900年，导致利息、年龄或截止日期计算出错。

各国政府和企业严阵以待，准备应对可能出现的混乱。1998年2月，克林顿总统成立委员会，协调美国的应对工作。顾问们审查旧系统，修补有问题的日期处理逻辑。1999年11月，美国商务部估计，美国各级政府和企业将为此花费约1000亿美元；对全球总支出的预测则高达数千亿美元。

1999年12月31日午夜来临，结果几乎没有出大问题。偶尔出现的小故障也大多轻微。日本志贺核电站的辐射监测设备失灵，但没有威胁安全；美国国防部一套处理间谍卫星数据的地面系统也出了故障。2000年2月，美国参议院的一个委员会认定，这些支出是合理的。

许多人却持不同看法，认为威胁被夸大了，甚至是个骗局。但局面之所以平稳，很大程度上正是因为大量问题已经修好。这就是“防备悖论”：预防措施发挥作用，使所抵御的危险没有出现，危险反而看起来像是凭空捏造的。繁琐的修复工作换来的不是感谢，而是嘲笑。

批评者指出，意大利和韩国花的钱少得多，结果却并不更差。但这种比较具有误导性：较低的支出数字往往没有算上企业的花费，而且很多修补工作也替换了早该升级的过时系统。没人能逐笔说清，哪些开支必不可少，哪些花得过多。奏效的预防可能只留下账单，人们只能拿它与那场本可能发生的灾难作比较。

文中的约1000亿美元是当时的估计；平稳过渡不代表每笔支出都已被证明必要。

核心词汇

先记这22项。连着下方的原文短语一起回想。

bug n.
程序漏洞
The Y2K bug

shortcut n.
捷径；省事的办法
began as a shortcut

ambiguity n.
歧义
That compact code hid an ambiguity

corrupt v.
使出错；破坏
corrupting calculations

brace for 动词短语
准备应对（困难）
firms braced for chaos

coordinate v.
协调
to coordinate America's effort

audit v.
审查；核查
Consultants audited legacy systems

legacy adj.
沿用的；旧有的
legacy systems

patch v.
修补（程序）
patched faulty date logic

projection n.
预测
projections of the global bill

sporadic adj.
零星的；偶发的
sporadic and mostly minor

expenditure n.
支出
the expenditure had been justified

justified adj.
合理的；有正当理由的
the expenditure had been justified

exaggerate v.
夸大
the threat had been exaggerated

hoax n.
骗局
even a hoax

preparedness paradox n.
防备悖论
the preparedness paradox

precaution n.
预防措施
when precautions work

ward off 动词短语
抵御
the danger they ward off

omit v.
遗漏
omitted what firms spent

obsolete adj.
过时的
replaced obsolete systems

essential adj.
必不可少的
which dollars were essential

catastrophe n.
灾难
a catastrophe that might have been

文中词汇与搭配

补充词汇共41项。保留每个表达在本文中的意思。

millennium n.
千年；千禧年

memory n.
计算机内存

hardware n.
硬件

costly adj.
昂贵的

programmer n.
程序员

digit n.
数字

compact adj.
精简紧凑的

code n.
程序代码

deadline n.
截止日期

chaos n.
混乱

president n.
总统

council n.
委员会

consultant n.
顾问

faulty adj.
有缺陷的

Commerce Department n.
美国商务部

enterprise n.
企业

global adj.
全球的

midnight n.
午夜

strike v.
（钟）敲响

glitch n.
小故障

文中词汇与搭配 续

补充词汇共41项。保留每个表达在本文中的意思。

crop up 动词短语
意外出现

minor adj.
轻微的

falter v.
失灵；运转不稳

spy n.
间谍

conclude v.
得出结论

prevail v.
占上风；普遍存在

largely adv.
很大程度上

imaginary adj.
想象出来的

tedious adj.
繁琐乏味的

mockery n.
嘲笑

gratitude n.
感激

South Korea n.
韩国

fare no worse 动词短语
情况并不更差

comparison n.
比较

mislead v.
误导

remedial adj.
补救的

overdue for 形容词短语
早该进行……的

excessive adj.
过多的；过度的

prevention n.
预防

invoice n.
账单

be weighed against 动词短语
与……权衡比较

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That compact code hid an _____: a machine might read "00" as 1900, corrupting calculations of interest, ages, or deadlines.

02 Consultants audited legacy systems and _____ faulty date logic.

03 Glitches that did crop up were _____ and mostly minor.

04 This is the preparedness paradox: when _____ work, the danger they ward off never appears, so it looks imaginary.

文章主线

- 隐患 旧程序为省内存只用两位数字记年，可能将2000年的“00”误读为1900年，算错利息、年龄和期限。
- 应对 政府与企业排查旧系统、修补日期逻辑；美国1999年预计花约1,000亿美元，全球预测达数千亿美元。
- 跨年 2000年跨年大体平稳，故障零星轻微；美国参议院委员会认定开支合理，却有人把威胁当成骗局。
- 悖论 平稳很大程度上来自预先修复：预防奏效、灾难未现，危险反倒像凭空捏造，这就是“防备悖论”。
- 限度 意大利、韩国的低支出统计常漏算企业花费，修补也更新了过时系统；究竟哪些钱必要、哪些过量，无法逐笔断定。

参考答案　1 ambiguity 2 patched 3 sporadic 4 precautions

被街道重塑的大脑

The Brain That Learned London



熟记伦敦街道，会怎样改变大脑？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Name two addresses and a London cabbie will rattle off the quickest route. That prowess is hard-won. To earn a licence, aspiring drivers must conquer "the Knowledge", an arduous examination on about 25,000 streets within six miles of Charing Cross. Hopefuls typically spend three to four years committing this labyrinth to memory, and only about half qualify.

A study published in PNAS in April 2000 scanned 16 male cabbies and 50 other men. The hippocampus is a seahorse-shaped structure vital to memory and navigation. In the cabbies, its rear, or posterior, portion was larger than in the other men, and its front, or anterior, portion smaller. Longer service went with a larger rear portion.

Yet the link left the cause uncertain. Perhaps an innate flair for spatial tasks lured people to the occupation, or perhaps stressful driving was the culprit. A December 2006 study dispelled the second notion: cabbies had more rear grey matter, the tissue dense with nerve cells, than bus drivers, confined to predetermined routes, of equivalent experience and stress.

A December 2011 longitudinal study tackled the first, scanning 79 trainees and 31 non-trainees at enrolment and three to four years later. Their brains began alike. Only the 39 who qualified exhibited an expansion of rear grey matter, so the sequence suggests learning preceded the enlargement.

The gain appeared to entail a trade-off: qualified drivers trailed the non-trainees on one test of recalling an intricate figure. The cohorts were small and exclusively male, yet the studies imply the healthy adult brain can remodel itself through intensive learning. Every street learned, it seems, occupied room once claimed by something else.

中文译文

只要说出两个地址，伦敦出租车司机就能迅速报出最快的路线。这份本领来之不易。为了取得执照，准司机必须攻克一项叫作“知识大考”的艰难考试，熟记距查令十字六英里以内约2.5万条街道。备考者通常花三到四年把这座街道迷宫牢记于心，最终只有约一半能够合格。

2000年4月发表于《美国国家科学院院刊》的一项研究，扫描了16名男性出租车司机和50名其他男性的大脑。海马体形似海马，对记忆和认路至关重要。与其他男性相比，出租车司机海马体的后部更大，前部则更小。从业时间越长，后部也越大。

然而，这种关联并不能确定原因。也许是天生擅长空间任务的人更容易被这个职业吸引，也许是紧张的驾驶工作造成了差异。2006年12月的一项研究排除了后一种解释：与驾驶经验、压力水平相当的公交司机相比，出租车司机海马体后部的灰质仍然更多。灰质是一种密布神经细胞的组织，而公交司机的行驶路线是预先固定的。

2011年12月的一项纵向研究开始检验前一种解释。研究者在79名学员报名时，以及三到四年后，分别扫描了他们和31名未参训者的大脑。各组起初的大脑状况相似，只有最终通过考试的39名学员，出现了海马体后部灰质增加。这一先后顺序提示，学习发生在灰质增大之前。

这种收获似乎也伴随着某种取舍：在一项回忆复杂图形的测试中，通过考试的司机表现不如未参训者。研究的样本群体较小，而且全部为男性，但结果仍提示，健康的成人大脑可以通过高强度学习重塑自身。看来，每记住一条街道，都像是占用了原先属于其他事物的一点空间。

核心词汇

先记这22项。连着下方的原文短语一起回想。

prowess n.
高超本领
That prowess is hard-won.

arduous adj.
艰难费力的
an arduous examination

commit v.
此处指把……牢记于心
committing this labyrinth to memory

qualify v.
取得资格；合格
and only about half qualify.

hippocampus n.
海马体
The hippocampus

navigation n.
辨路；导航
vital to memory and navigation.

posterior adj.
后部的
its rear, or posterior, portion

anterior adj.
前部的
its front, or anterior, portion

uncertain adj.
无法确定的
Yet the link left the cause uncertain.

innate adj.
天生的
an innate flair for spatial tasks

spatial adj.
空间的
spatial tasks

dispel v.
消除；打消
study dispelled the second notion:

predetermined adj.
预先确定的
confined to predetermined routes

equivalent adj.
相当的；等同的
of equivalent experience and stress.

longitudinal adj.
长期追踪同一群体的
longitudinal study

expansion n.
扩大；增加
an expansion of rear grey matter

precede v.
发生在……之前
learning preceded the enlargement.

entail v.
意味着；必然涉及
to entail a trade-off:

trade-off n.
取舍；有得有失
appeared to entail a trade-off

cohort n.
接受研究的一组人
The cohorts were small

imply v.
提示；意味着
yet the studies imply

remodel v.
重塑；重新构建
can remodel itself

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

rattle off phr. v.
快速流利地说出

hard-won adj.
来之不易的

licence n.
执照；许可证

aspiring adj.
有志于成为……的

conquer v.
攻克；征服

examination n.
考试

hopeful n.
有望成功者；此处指备考者

typically adv.
通常

labyrinth n.
迷宫

scan v.
扫描检查

vital adj.
至关重要的

portion n.
部分

flair n.
天赋；才能

lure v.
吸引；诱使

occupation n.
职业

stressful adj.
造成精神压力的

culprit n.
问题的起因；罪魁祸首

notion n.
看法；想法

dense adj.
密集的

nerve n.
神经

confine v.
限制；使局限于

tackle v.
着手处理；检验

trainee n.
受训学员

enrolment n.
报名；入学

exhibit v.
表现出；显示

sequence n.
先后顺序

enlargement n.
增大

trail v.
落后于

intricate adj.
复杂精细的

exclusively adv.
全部；仅仅

intensive adj.
高强度的；集中的

occupy v.
占据

claim v.
此处指占用

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 To earn a licence, aspiring drivers must conquer "the Knowledge", an _____ examination on about 25,000 streets within six miles of Charing Cross.

02 The _____ is a seahorse-shaped structure vital to memory and navigation.

03 Yet the link left the cause _____.

04 The gain appeared to entail a _____: qualified drivers trailed the non-trainees on one test of recalling an intricate figure.

文章主线

- 训练 伦敦出租车司机通常花三四年记住约2.5万条街道，只有约一半学员通过考试。
- 差异 海马体参与记忆与认路；司机的海马体后部比对照男性大、前部小，最初无法判断原因。
- 排查 与经验、压力相近但走固定路线的公交司机相比，出租车司机的海马体后部灰质仍更多。
- 追踪 学员与对照者起初脑部相似；三四年后，仅考过者海马体后部灰质增加，支持成人大脑可随学习改变。
- 边界 样本小且全为男性；考过者在一项图形记忆测试中比未参训者差，提示认路能力增强可能伴随记忆取舍。

参考答案　1 arduous 2 hippocampus 3 uncertain 4 trade-off

两次卖掉埃菲尔铁塔的骗子

The Man Who Sold the Eiffel Tower Twice



铁塔怎么成了骗子的“货物”？

英文原文

In 1925 six scrap-metal traders met secretly at the Hôtel de Crillon in Paris. A deputy director-general of the Ministry of Posts and Telegraphs shared the momentous news, confidentially, that the Eiffel Tower was to be sold for scrap. He invited tenders and made each dealer solemnly vow secrecy, ostensibly to spare the public indignation.

The pretext was not absurd. Built for the 1889 World's Fair, the monument had originally been destined to come down after twenty years. The high-ranking deputy was Victor Lustig, a crook whose ministerial stationery was a forgery. He had read a newspaper article in which the authorities lamented the hefty cost of the tower's upkeep.

André Poisson, a businessman impatient to gain a foothold in Paris, won. Lustig intimated that his modest civil-service remuneration needed supplementing, and Poisson paid an additional bribe. Lustig left for Vienna with the takings.

Fleeced and too ashamed to go to the police, Poisson swallowed the loss quietly. Not a whisper of the deceit appeared in print.

About a month later Lustig had the gall to reappear with an encore of his script for new dealers. This time the winner grew distrustful and went to the police. The affair made the papers, and Lustig made his getaway. The much-repeated second sale is a fairy tale.

Later arrested in the United States for making fake money, Lustig broke out of detention, was recaptured 27 days later and served time on the penal island of Alcatraz. He died of pneumonia in 1947.

Of Lustig's two marks, the one who paid kept mum and the one who kept his money went to the police.

中文译文

1925年，六名废金属商人在巴黎克里雍酒店秘密会面。邮电部的一名副总负责人私下告诉他们一则重大消息：埃菲尔铁塔将当作废铁出售。他邀请商人投标，并要求每个人郑重发誓保密，表面上的理由是避免引起公愤。

这个借口并不荒唐。铁塔为1889年世博会而建，原本计划二十年后拆除。那位位高权重的“副总负责人”其实是骗子维克多·卢斯蒂格，带有政府部门抬头的信笺也是伪造的。他曾在报上看到，当局抱怨铁塔维护费用高昂。

急于在巴黎站稳脚跟的商人安德烈·普瓦松中标。卢斯蒂格暗示，自己微薄的公务员薪酬需要补贴，普瓦松便又付了一笔贿赂。随后，卢斯蒂格带着到手的钱前往维也纳。

普瓦松被骗了钱，却因羞于报警而默默咽下损失。报纸上连关于这场骗局的一点风声都没有。

大约一个月后，卢斯蒂格竟又厚着脸皮回来，向新一批商人重演旧把戏。这一次，中标者起了疑心，前去报警。事情登上报纸，卢斯蒂格逃之夭夭。广为流传的“第二次卖成铁塔”，其实只是虚构故事。

后来，卢斯蒂格在美国因制造假币被捕。他从拘留处逃出，27天后再次落网，并在恶魔岛监狱服刑。1947年，他死于肺炎。

卢斯蒂格的两个行骗对象中，付了钱的那人守口如瓶，保住钱的那人反而报了警。

标题里的“两次卖掉”是传说；原文明确，第二次骗局没有得手。

核心词汇

先记这22项。连着下方的原文短语一起回想。

scrap-metal adj.
废金属的
six scrap-metal traders

confidentially adv.
私下地；保密地
confidentially

tender n.
投标
He invited tenders

vow v.
发誓；郑重保证
solemnly vow secrecy

ostensibly adv.
表面上；声称地
ostensibly to spare

public indignation 名词短语
公愤；公众愤怒
the public indignation

pretext n.
借口；托词
The pretext was not absurd

destined adj.
原定的；注定的
destined to come down

crook n.
骗子；不法之徒
a crook

stationery n.
信笺；办公文具
ministerial stationery

forgery n.
伪造品；伪造文件
was a forgery

upkeep n.
维护；保养
the tower's upkeep

gain a foothold 动词短语
站稳脚跟；取得一席之地
gain a foothold in Paris

intimate v.
暗示
Lustig intimated

remuneration n.
报酬；薪酬
civil-service remuneration

bribe n.
贿赂；贿款
an additional bribe

fleece v.
骗取钱财；敲竹杠
Fleeced and too ashamed

deceit n.
欺骗；骗局
a whisper of the deceit

have the gall to 动词短语
竟敢厚着脸皮做……
had the gall to reappear

grow distrustful 动词短语
起疑；变得不信任
the winner grew distrustful

detention n.
拘留；羁押
broke out of detention

keep mum 动词短语
守口如瓶；保持沉默
the one who paid kept mum

文中词汇与搭配

补充词汇共35项。保留每个表达在本文中的意思。

trader n.
商人；交易者

secretly adv.
秘密地

deputy adj.
副职的

director-general n.
总负责人；总局长

momentous adj.
重大的；影响深远的

solemnly adv.
郑重地；严肃地

secrecy n.
保密；秘密状态

absurd adj.
荒唐的

monument n.
纪念建筑；纪念物

originally adv.
原本；最初

come down 动词短语
被拆除；倒下

high-ranking adj.
高级别的；位高权重的

ministerial adj.
政府部门的；部级机关的

lament v.
哀叹；抱怨

hefty cost 名词短语
高昂费用

businessman n.
商人

impatient adj.
急切的；迫不及待的

civil-service adj.
公务员的；文官的

文中词汇与搭配 续

补充词汇共35项。保留每个表达在本文中的意思。

supplement v.
补充；补贴

additional adj.
额外的

takings n.
到手的钱；收入

ashamed adj.
羞愧的；感到丢人的

quietly adv.
默默地；不声张地

whisper n.
一点风声；耳语

reappear v.
再次出现

encore n.
重演；再来一遍

script n.
台词；既定说辞

affair n.
事件；事情

getaway n.
逃脱；逃走

much-repeated adj.
被反复讲述的

fairy tale 名词短语
童话；虚构故事

fake adj.
假的；伪造的

recapture v.
再次抓获

serve time 动词短语
服刑

penal adj.
刑罚的；监禁的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ was not absurd.

02 The high-ranking deputy was Victor Lustig, a crook whose ministerial stationery was a _____.

03 Lustig intimated that his modest civil-service _____ needed supplementing, and Poisson paid an additional bribe.

04 Of Lustig's two marks, the one who paid _____ and the one who kept his money went to the police.

文章主线

- 冒充 1925年，卢斯蒂格伪造公文、冒充邮电部官员，邀六名废金属商秘密投标，谎称埃菲尔铁塔要卖废铁。
- 借口 铁塔原拟在世博会20年后拆除，维护费又高；卢斯蒂格利用这些真背景，并以免引公愤为由要求保密。
- 得手 急于在巴黎立足的普瓦松付了购塔款和贿赂，受骗后却羞于报警，让卢斯蒂格带钱逃走。
- 再骗 约一月后，卢斯蒂格重演骗局，新买家起疑报警且未付款；所谓第二次卖成铁塔的故事并不成立。
- 下场 卢斯蒂格后来在美国因造假币被捕，越狱27天后再落网，在恶魔岛监狱服刑，1947年死于肺炎。

参考答案　1 pretext 2 forgery 3 remuneration 4 kept mum

罗马混凝土的千年之谜

The Secret of Rome's Undying Concrete



古罗马混凝土为何如此耐久？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Nearly 1,900 years after its completion, in about AD 125–128, the Pantheon's dome still crowns the vast chamber below. About 43 metres in diameter, it remains the world's largest unreinforced concrete dome. It is an astonishing feat of architecture, and a stubborn puzzle.

Modern concrete is usually reinforced with embedded steel bars, which are prone to corrosion. As the steel rusts, it swells and fractures the concrete around it, and many structures are designed for a lifespan of around 50 years. Roman concrete is not stronger than its modern counterpart; its merit lies in durability.

Yet the ancient material seemed to have a shortcoming. It is dotted with white chunks of lime, known as lime clasts. Experts long ascribed these lumps to sloppy mixing or inferior raw ingredients, a defect that, they assumed, a more conscientious builder would have avoided.

A study published in Science Advances in January 2023 offered a different interpretation. The clasts, it suggested, were a by-product of "hot mixing" with quicklime, a reactive lime that heats up in water. When water seeps into a crack, the clasts dissolve and their calcium recrystallises as calcium carbonate, plugging the gap. In tests, water ceased to trickle through cracked hot-mixed samples within weeks, but kept flowing through samples without quicklime.

The implications extend beyond archaeology. Cement manufacture accounts for about 8 per cent of global carbon dioxide emissions, the think-tank Chatham House reported in June 2018. Concrete that mends its cracks could last longer, making demolition and rebuilding rarer and trimming emissions. The white lumps once scorned as signs of clumsy workmanship have prompted a rethink of construction.

中文译文

万神殿约在公元125至128年竣工。近1900年过去，它的穹顶依然高踞在宽阔的殿堂之上。穹顶直径约43米，至今仍是世界上最大的无钢筋混凝土穹顶。这是建筑史上的惊人壮举，也是一道始终难解的谜题。

现代混凝土通常用埋入其中的钢筋加固，而钢筋容易锈蚀。钢筋生锈后会膨胀，撑裂周围的混凝土；许多建筑的设计寿命约为50年。古罗马混凝土的强度并不比现代混凝土更高，它的优势在于耐久。

不过，古代材料看起来也有一个缺点：内部散布着白色石灰小块，称为石灰碎屑。专家长期把这些团块归咎于搅拌马虎或原料劣质，认为这是一种更认真负责的工匠本应避免的缺陷。

2023年1月，《科学进展》的一项研究提出了另一种解释。研究认为，这些碎屑是使用生石灰进行“热混合”留下的副产物；生石灰活性强，遇水会放热。水渗入裂缝时，碎屑会溶解，其中的钙重新结晶为碳酸钙，堵住缝隙。试验中，开裂的热混合样本在几周内就不再渗水，而不含生石灰的样本仍持续漏水。

这一发现的意义不止于考古。查塔姆研究所2018年6月报告称，水泥生产约占全球二氧化碳排放的8%。能自行修补裂缝的混凝土，可能延长建筑寿命，减少拆除与重建，进而降低排放。那些曾被轻视为手艺拙劣证据的白色团块，如今促使人们重新思考建造方式。

核心词汇

先记这22项。连着下方的原文短语一起回想。

feat n.
壮举；非凡成就
It is an astonishing feat of

prone to adj.phr.
容易发生；易于
which are prone to corrosion.

corrosion n.
锈蚀；腐蚀
which are prone to corrosion.

lifespan n.
使用寿命
for a lifespan of around 50 years.

counterpart n.
对应的事物
stronger than its modern counterpart;

durability n.
耐久性
its merit lies in durability.

shortcoming n.
缺点；不足
material seemed to have a shortcoming.

ascribe to phr.v.
与to连用：归因于
Experts long ascribed these lumps to

inferior adj.
劣质的
mixing or inferior raw ingredients,

conscientious adj.
认真负责的
more conscientious builder would have

interpretation n.
解释
offered a different interpretation.

by-product n.
副产物
were a by-product of "hot mixing" with

quicklime n.
生石灰
of "hot mixing" with quicklime,

reactive adj.
化学活性强的
reactive lime that heats up in water.

seep into phr.v.
缓慢渗入
When water seeps into a crack,

dissolve v.
溶解
the clasts dissolve and their calcium

implication n.
意义；可能影响
The implications extend beyond

extend beyond v.phr.
超出……范围
The implications extend beyond

account for phr.v.
占……比例
Cement manufacture accounts for about

emission n.
排放量
of global carbon dioxide emissions,

demolition n.
拆除
making demolition and rebuilding rarer

prompt v.
促使
have prompted a rethink of

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

completion n.
竣工；完成

dome n.
穹顶

crown v.
位于……顶部

vast adj.
宽广的；巨大的

chamber n.
大厅；殿堂

in diameter prep.phr.
直径为

unreinforced concrete n.phr.
无钢筋混凝土

astonishing adj.
令人惊叹的

architecture n.
建筑；建筑艺术

stubborn adj.
难以解决的

puzzle n.
谜题

reinforce with v.phr.
用……加固

embedded adj.
埋入内部的

steel bar n.phr.
钢筋

rust v.
生锈

swell v.
膨胀

fracture v.
使断裂；使开裂

structure n.
建筑物；结构

merit n.
优点；长处

dotted with adj.phr.
散布着

chunk n.
小块；厚块

clast n.
碎屑；碎屑颗粒

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

lump n.
团块

to prep.
标示所归因的对象

sloppy adj.
马虎的

raw ingredient n.phr.
原材料

defect n.
缺陷

assume v.
认为；未经证实地假定

heat up phr.v.
变热；升温

crack n.
裂缝

calcium n.
钙

plug v.
堵住

gap n.
缝隙

cease to v.phr.
不再；停止

trickle v.
细细流淌

within weeks prep.phr.
在几周之内

archaeology n.
考古学

manufacture n.
生产；制造

think-tank n.
智囊团；研究机构

mend v.
修补

trim v.
削减

scorn v.
轻视

clumsy adj.
拙劣的；笨拙的

rethink n.
重新考虑

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 which are prone to _____.

02 its merit lies in _____.

03 When water _____ a crack,

04 Cement manufacture _____ about 8 per cent of global carbon dioxide emissions,

文章主线

- 耐久 万神殿直径约43米的无钢筋穹顶已存近1900年；罗马混凝土长于耐久，并非强度高于现代混凝土。
- 疑点 现代钢筋锈胀会撑裂混凝土；罗马材料中的白色石灰块，过去却被当成搅拌差或原料劣的缺陷。
- 新解 2023年研究认为，生石灰热混合留下石灰块；水渗入裂缝后，石灰块溶解、钙重新结晶为碳酸钙，堵住裂缝。
- 试验 开裂的热混合样本几周内不再渗水，未加生石灰的对照仍漏水，支持这种自愈解释。
- 前景 水泥生产约占全球二氧化碳排放8%；若自愈材料能延长建筑寿命，就有望减少拆建及排放。

参考答案　1 corrosion 2 durability 3 seeps into 4 accounts for

一夜之间换边开车

The Night Sweden Switched Sides



全国换边行驶，怎样在一个清晨完成?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In a 1955 referendum, about 83 per cent of voters chose to keep driving on the left, an overwhelmingly clear rejection of change. On 10 May 1963, however, legislators in parliament overruled the electorate, approving right-hand traffic from 1967 in defiance of a populace among whom the plan was unpopular.

The case rested on conformity. Norway and Finland drove on the right, with about five million border crossings a year, so a left-hand orientation meant isolation and inconvenient journeys. Safety favoured alignment too. About 90 per cent of cars were left-hand drive, so motorists overtaking on narrow two-lane highways risked head-on collisions.

A state body ran the mammoth preparation under a logo, a large H with an arrow. Its propaganda included a song contest on television, and the winning jingle, whose lyrics urged people to keep right, reached fifth in the chart. Some 360,000 items of signage were changed, and a massive fleet of over 1,000 new buses with right-hand doors was procured.

The transition came on Sunday 3 September 1967. A curfew kept non-essential vehicles off the asphalt from one to six, with essential traffic exempt. At ten to five, all traffic stopped and crawled across, restarting at five, before dawn, with its direction flipped. Stockholm and Malmö had a longer closure, from Saturday morning to Sunday afternoon, while crews reconfigured junctions.

On the following Monday, 125 accidents were reported, against 130 to 198 on earlier Mondays, a statistical improvement put down to the heightened care of hesitant drivers. The nervousness apparently subsided and prudent habits faded, for by 1969 accident and fatality rates had reverted to their baseline.

中文译文

1955年瑞典举行公投，约83%的投票者选择保留靠左行驶，以压倒性多数拒绝改变。然而，1963年5月10日，议会推翻选民的选择，不顾这项计划在民众中缺乏支持，批准自1967年起改为靠右行驶。

改制的理由是与邻国统一。挪威和芬兰都靠右行驶，每年约有500万次跨境通行，瑞典靠左便带来隔阂和出行不便。安全也支持统一：约90%的汽车方向盘位于左侧，司机在狭窄的双车道公路上超车时，容易面临迎面相撞的危险。

一个国家机构负责这项庞大的准备工程，以带箭头的大写H为标识。宣传还包括电视歌曲比赛，获胜的广告歌用歌词劝人靠右行驶，甚至登上歌曲排行榜第五名。约36万件道路标识被更换，1000多辆右侧开门的新公交车也采购到位。

切换在1967年9月3日星期日进行。凌晨1点至6点，非必要车辆不得上路，必要车辆获得豁免。4点50分，仍在路上的车辆全部停下，再缓缓移到另一侧；5点、天亮之前，车辆统一按相反的行驶侧重新出发。斯德哥尔摩和马尔默的封路时间更长，从周六上午到周日下午，施工人员在此期间改造交叉路口。

随后的星期一报告了125起事故，此前的星期一则有130至198起。这一统计上的改善，被归因于司机对新制度尚不熟悉，开车格外小心。紧张感似乎后来消退了，谨慎习惯也随之淡去：到1969年，事故率和死亡率已回到原来的基准水平。

左侧方向盘与靠左行驶不是一回事；首个周一事故少，并不意味着长期事故率持续下降。

核心词汇

先记这22项。连着下方的原文短语一起回想。

referendum n.
全民公投
a 1955 referendum

legislator n.
立法者
legislators in parliament

overrule v.
推翻（决定或意见）
overruled the electorate

electorate n.
全体选民
overruled the electorate

in defiance of 介词短语
不顾；违抗
in defiance of a populace

conformity n.
一致；统一
The case rested on conformity

isolation n.
孤立
meant isolation

alignment n.
统一；保持一致
Safety favoured alignment too

overtake v.
超车
motorists overtaking

head-on collision n.
迎头相撞
risked head-on collisions

preparation n.
准备工作
the mammoth preparation

signage n.
各类标识
360,000 items of signage

procure v.
采购；取得
was procured

transition n.
转变；过渡
The transition came on Sunday

exempt adj.
获豁免的
with essential traffic exempt

reconfigure v.
重新设置；改造
crews reconfigured junctions

statistical adj.
统计上的
a statistical improvement

put … down to … 动词短语
把……归因于……
put down to the heightened care

subside v.
消退
The nervousness apparently subsided

prudent adj.
谨慎的
prudent habits faded

revert to 动词短语
回到；恢复到
had reverted to their baseline

baseline n.
基准水平
reverted to their baseline

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

overwhelmingly adv.
压倒性地

rejection n.
拒绝

parliament n.
议会

populace n.
民众

unpopular adj.
不得人心的

orientation n.
行驶侧；通行取向

inconvenient adj.
不便的

motorist n.
驾车者

lane n.
车道

highway n.
公路

mammoth adj.
庞大的

logo n.
标识

arrow n.
箭头

propaganda n.
宣传

contest n.
比赛

jingle n.
广告短歌

lyric n.
歌词

chart n.
歌曲排行榜

massive adj.
巨大的

fleet n.
车队

curfew n.
车辆限行令

non-essential adj.
非必要的

asphalt n.
沥青路面

crawl v.
缓慢移动

dawn n.
黎明

flip v.
翻转；改变方向

closure n.
封闭

junction n.
交叉路口

improvement n.
改善

heightened adj.
增强的

hesitant adj.
犹豫不决的

nervousness n.
紧张

fade v.
逐渐消失

fatality rate n.
死亡率

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In a 1955 _____, about 83 per cent of voters chose to keep driving on the left, an overwhelmingly clear rejection of change.

02 Some 360,000 items of _____ were changed, and a massive fleet of over 1,000 new buses with right-hand doors was procured.

03 A curfew kept non-essential vehicles off the asphalt from one to six, with essential traffic _____.

04 The nervousness apparently subsided and prudent habits faded, for by 1969 accident and fatality rates had reverted to their _____.

文章主线

- 缘由 瑞典靠左，挪威和芬兰靠右，过境车流频繁；瑞典九成汽车的方向盘又在左边，窄路超车有迎面相撞风险。
- 决定 1955年公投约83%反对改靠右，议会仍于1963年批准改制，定在1967年实施。
- 准备 政府广泛宣传，更换约36万件道路标识，购置1,000多辆右侧开门的公交车，并在切换期间限制非必要车辆上路。
- 切换 1967年9月3日4点50分，车辆停下后缓缓移到另一侧，5点统一靠右出发；部分城市另封路改造路口。
- 后续 切换后首个周一事故减少，被归因于司机格外谨慎；但到1969年，事故率和死亡率都回到原有水平。

参考答案　1 referendum 2 signage 3 exempt 4 baseline

微塑料：
无处不在，危害几何?

Microplastics: Everywhere, but How Harmful?



检出了微塑料，就能证明它有害吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In May 2004 Richard Thompson, a marine biologist at the University of Plymouth, and colleagues described an overlooked pollutant in the journal Science. Sifting through plankton samples kept since the 1960s, they found that microscopic plastic fragments and synthetic fibres had grown more abundant. They labelled the particles microplastics.

Since then the durable debris has proved ubiquitous, turning up on shorelines, in deep-sea sediment, in rainfall, and in Arctic ice. Regulation came before proof of harm to people: in December 2015 America passed a law banning plastic microbeads from scrubs and other cosmetics rinsed down drains.

The search turned inward. In March 2022 a small Dutch study found plastic in the blood of about three-quarters of 22 healthy donors. In March 2024 a study in the New England Journal of Medicine linked plastic in the fatty plaque of neck arteries to a higher risk of heart attack, stroke, or death. Its authors cautioned that correlation is not causation.

The most sensational figures are the shakiest. In June 2019 a report for WWF, a conservation group, reckoned people might swallow a credit card's worth of plastic a week; critics call that an overestimate. Two months later the World Health Organization was reassuring but tentative: on limited evidence, microplastics in drinking water posed no apparent risk at current levels. In February 2025 a study in Nature Medicine reported plastic in brain tissue in an amount popularised as a spoonful; others queried its methodology.

Detection has outrun understanding; human studies remain small or observational. Harm is plausible but unproven, and until the verdict is in, better measurement may guide policy more safely than panic.

中文译文

2004年5月，普利茅斯大学海洋生物学家理查德·汤普森和同事在《科学》杂志上描述了一种长期被忽略的污染物。他们细查自20世纪60年代起保存的浮游生物样本，发现其中微小的塑料碎片和合成纤维越来越多。他们把这些颗粒称为微塑料。

此后，人们发现，这些难以消失的碎屑几乎无处不在：海岸、深海沉积物、雨水和北极冰层中都有它们的踪迹。监管先于人体危害的确证出现：2015年12月，美国通过法律，禁止在磨砂膏等用后冲入下水道的化妆品中添加塑料微珠。

寻找微塑料的目光随后转向人体内部。2022年3月，一项小规模荷兰研究在22名健康献血者中，发现约四分之三的人血液里有塑料。2024年3月，《新英格兰医学杂志》发表的一项研究发现，颈部动脉脂肪性斑块中的塑料与较高的心脏病发作、中风或死亡风险相关。作者提醒：相关不等于因果。

最耸人听闻的数字，往往也最站不住脚。2019年6月，一份为自然保护组织WWF撰写的报告估计，人每周可能吞下相当于一张信用卡重量的塑料；批评者认为这个数字过高。两个月后，世界卫生组织给出了令人稍感安心、却仍属初步的判断：依据有限证据，饮用水中的微塑料在当时的含量下并无明显风险。2025年2月，《自然·医学》上的一项研究报告了脑组织中的塑料，其数量被大众转述成“一勺”；另一些人则质疑这项研究的方法。

检测走在了理解前面；人体研究仍然规模较小，或只属于观察性研究。微塑料可能有害，但尚未得到证实。在结论明确之前，用更准确的测量来指导政策，或许比任由恐慌主导更稳妥。

核心词汇

先记这22项。连着下方的原文短语一起回想。

pollutant n.
污染物
described an overlooked pollutant in

sift through phr.v.
细查；逐一筛查
Sifting through plankton samples kept

microscopic adj.
极微小的；显微镜下的
that microscopic plastic fragments and

synthetic adj.
合成的；人造的
plastic fragments and synthetic fibres

abundant adj.
大量存在的
fibres had grown more abundant.

durable adj.
耐久的；不易分解的
Since then the durable debris has

debris n.
碎屑；残片
Since then the durable debris has

ubiquitous adj.
无处不在的
durable debris has proved ubiquitous,

sediment n.
沉积物
on shorelines, in deep-sea sediment,

regulation n.
监管；规管
Regulation came before proof of harm

donor n.
献血者
three-quarters of 22 healthy donors.

artery n.
动脉
in the fatty plaque of neck arteries

correlation n.
相关性
Its authors cautioned that correlation

causation n.
因果关系
that correlation is not causation.

sensational adj.
耸人听闻的
The most sensational figures are the

overestimate n.
过高的估计
critics call that an overestimate.

tentative adj.
初步的；未定的
was reassuring but tentative:

methodology n.
研究方法；方法体系
others queried its methodology.

observational adj.
观察性的
studies remain small or observational.

plausible adj.
有可能的；合乎情理的
Harm is plausible but unproven,

verdict n.
定论；判断
and until the verdict is in, better

measurement n.
测量
is in, better measurement may guide

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

marine adj.
海洋的

biologist n.
生物学家

plankton n.
浮游生物

fragment n.
碎片

fibre n.
纤维

label v.
把……称为；命名

particle n.
颗粒；微粒

turn up phr.v.
出现；被发现

shoreline n.
海岸线

rainfall n.
降雨；雨水

pass a law v.phr.
通过一项法律

scrub n.
磨砂清洁产品

cosmetic n.
化妆品

rinse down phr.v.
随冲洗流入

turn inward v.phr.
转向内部；此处指人体内

fatty adj.
含脂肪的

plaque n.
血管内的斑块

stroke n.
中风

caution v.
告诫；提醒谨慎

shaky adj.
最不可靠的

conservation n.
自然保护

reckon v.
估计；估算

swallow v.
吞下

reassuring adj.
令人安心的

apparent adj.
明显的

tissue n.
生物组织

spoonful n.
一勺的量

query v.
质疑

detection n.
检测；发现

unproven adj.
尚未证实的

panic n.
恐慌

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 _____ plankton samples kept since the 1960s,

02 Its authors cautioned that _____ is not causation.

03 Two months later the World Health Organization was reassuring but _____:

04 Harm is _____ but unproven,

文章主线

- 发现 2004年，汤普森团队在存档海洋样本中发现微小塑料碎片和合成纤维增多，将其称为微塑料。
- 扩散 微塑料从深海到北极无处不在；美国2015年立法禁用冲洗类化妆品塑料微珠，当时人体危害尚未证实。
- 证据 22名健康献血者中，约四分之三的人血中检出塑料；另一研究发现颈部动脉斑块中的塑料与更高的心梗、中风或死亡风险相关，不能据此证明因果。
- 争议 每周吞下信用卡分量塑料的估算被批高估；脑内塑料量被转述成“一勺”，研究方法受质疑。
- 判断 世卫2019年据有限证据，初判饮水微塑料在当时含量下无明显风险；人体危害仍未定论。

参考答案　1 Sifting through 2 correlation 3 tentative 4 plausible

冰里的五千年凶案

The 5,000-Year-Old Murder in the Ice



冰人的箭伤，揭开了怎样的古老谜案?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 19 September 1991 two German hikers on a trek in the Ötztal Alps found a corpse by a glacier. It lay on a ridge at about 3,210 metres, on the border between Austria and Italy. The assumption was that a modern-day climber had died on an alpine slope, since glaciers sometimes yield frozen remains.

Taking it for a recent death, rescuers used a jackhammer that ripped flesh from his left hip. His possessions told another story: leather and grass clothing, a copper axe, a bow and arrows, all prehistoric artefacts. Known as Ötzi, the natural mummy was a man who lived about 5,300 years ago, between 3350 and 3105 BC.

In 2001, X-rays revealed a flint arrowhead lodged in his left shoulder, unseen by examiners a decade earlier. A scanner showed that the puncture had ruptured an artery, a mortal injury causing heavy bleeding, and left a large clot. Researchers believe he died within minutes. He was about 46.

Another person, it is thought, killed him. From the wound's depth researchers inferred that an archer shot him from the rear, at about 30 metres. The shooter probably caught him unawares. A 2018 study found he had dined on dried ibex meat and fat, red deer, and einkorn wheat, a hearty, nourishing meal reportedly consumed within two hours of his death.

Curators in Bolzano now keep Ötzi, whose 61 tattoos on joints and spine were perhaps an early form of acupuncture. The archer's identity is an enigma, and any motive is conjecture. Ötzi has a name, a last meal, and a cause of death, while his anonymous killer has none.

中文译文

1991年9月19日，两名德国徒步者在奥茨塔尔阿尔卑斯山远足时，于冰川旁发现一具遗体。它位于奥地利与意大利交界处、海拔约3210米的山脊上。人们最初以为，一名现代登山者死在高山坡地上，毕竟冰川有时会吐露冰冻的遗骸。

救援人员以为这是近期的死亡事件，使用风钻凿冰，结果撕损了死者左侧髋部的皮肉。但他的随身物品讲述了另一种故事：皮革和草编衣物、铜斧、弓箭，全是史前器物。这具被称为奥茨的天然木乃伊是一名男子，生活于约5300年前，即公元前3350至前3105年之间。

2001年，X光检查发现一枚燧石箭头嵌在他的左肩里，十年前的检查者并未发现它。扫描显示，穿刺伤撕裂了一条动脉，造成大量出血的致命损伤，并留下一个大血块。研究者认为，他在几分钟内便死亡，当时约46岁。

人们认为，他是遭人杀害的。研究者根据伤口深度推断，弓箭手从他背后约30米处射箭，很可能是在他毫无防备时出手。2018年的研究发现，他最后吃过干野山羊肉和脂肪、马鹿肉与单粒小麦。这是一顿丰盛而营养丰富的饭，据报道在死前两小时内吃下。

如今，博尔扎诺的馆藏保管人员保存着奥茨的遗体。他关节和脊柱上的61处文身，也许是一种早期针灸形式。弓箭手的身份仍是谜团，任何作案动机都只是猜测。奥茨有名字、最后一餐和死因；对于那名身份不明的凶手，我们却连这些信息都不知道。

核心词汇

先记这22项。连着下方的原文短语一起回想。

glacier n.
冰川
a corpse by a glacier

ridge n.
山脊
on a ridge at about 3,210 metres

assumption n.
假定；最初的判断
The assumption was

possession n.
随身物品；所有物
His possessions told another story

prehistoric adj.
史前的
all prehistoric artefacts

artefact n.
人工制品；考古器物
all prehistoric artefacts

mummy n.
木乃伊；保存下来的干尸
the natural mummy

flint n.
燧石
a flint arrowhead

lodge v.
嵌入；卡住
lodged in his left shoulder

puncture n.
穿刺伤；刺孔
the puncture had ruptured an artery

rupture v.
使破裂；撕裂
had ruptured an artery

mortal adj.
致命的
a mortal injury

infer v.
推断；推知
researchers inferred

unawares adv.
在没有防备时
caught him unawares

nourishing adj.
有营养的
a hearty, nourishing meal

curator n.
馆藏保管人员；策展人
Curators in Bolzano

acupuncture n.
针灸
an early form of acupuncture

identity n.
身份
The archer's identity

enigma n.
谜团
an enigma

motive n.
动机
any motive is conjecture

conjecture n.
猜测；推测
any motive is conjecture

anonymous adj.
身份不明的；不知姓名的
his anonymous killer

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

hiker n.
徒步者

trek n.
长途徒步；远足

corpse n.
尸体；遗体

border n.
边界；交界处

modern-day adj.
现代的；当代的

climber n.
登山者

alpine adj.
高山的；阿尔卑斯山的

slope n.
山坡；斜坡

jackhammer n.
风钻；气动破碎工具

rip v.
撕裂；扯下

flesh n.
皮肉；肌肉组织

hip n.
髋部

leather n.
皮革

copper adj.
铜制的

axe n.
斧子

unseen adj.
未被看见或发现的

by prep.
被；由

examiner n.
检查人员

scanner n.
扫描仪

bleeding n.
出血

clot n.
血块

it is thought 插入语
据认为；人们推测

archer n.
弓箭手

rear n.
后方；背后

shooter n.
射手

catch v.
使措手不及；趁其不备

dine on 动词短语
吃；进食

red deer 名词短语
马鹿

hearty adj.
丰盛的；充足的

consume v.
食用；吃下

tattoo n.
文身

spine n.
脊柱

killer n.
凶手

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ was that a modern-day climber had died on an alpine slope, since glaciers sometimes yield frozen remains.

02 From the wound's depth researchers _____ that an archer shot him from the rear, at about 30 metres.

03 Curators in Bolzano now keep Ötzi, whose 61 tattoos on joints and spine were perhaps an early form of _____.

04 The archer's identity is an enigma, and any motive is _____.

文章主线

- 误判 1991年阿尔卑斯山发现冰尸，最初被当作现代登山者，救援风钻还损伤了他的髋部。
- 身份 皮革和草编衣物、铜斧与弓箭显示他来自史前；冰人奥茨生活于约5300年前，死时约46岁。
- 死因 2001年发现肩内箭头，扫描显示动脉被撕裂；研究者认为他因此大量失血，数分钟内死亡。
- 重建 据伤口推测，射手从背后约30米放箭；胃中有野山羊肉和脂肪、鹿肉与小麦，据报道在死前两小时内吃下。
- 未解 关节和脊柱上的61处文身，或与早期针灸有关；但射手是谁、为何杀他，仍没有定论。

参考答案　1 assumption 2 inferred 3 acupuncture 4 conjecture

一颗棉花糖的真相

The Truth About One Marshmallow



一颗棉花糖能预测孩子的未来吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Picture a nursery school on Stanford's campus in the late 1960s and early 1970s. A child of about four or five sits in solitude before one treat, facing a blunt bargain: eat it now, or resist temptation for about 15 minutes and earn two. Impulsive children often surrender instantly; resolute ones hold out.

Follow-ups made the test legendary. A 1990 study in Developmental Psychology found that, as adolescents, longer waiters tended to attain higher scores on the SAT, a US college-entrance exam. A trivial snack became an emblem of virtue, and restraint at four was glorified as a predictor of destiny.

Yet the foundation was precarious. Only a few dozen children entered the SAT analysis, predominantly the sons and daughters of Stanford faculty and postgraduate students. Such a homogeneous, privileged group could scarcely represent ordinary families, and prosperous homes may foster patience and academic success alike.

A study published in Psychological Science in May 2018 ran a conceptual replication: the same idea, a new sample, modified methods. More than 900 youngsters from diverse backgrounds were assessed at four and a half and again at 15. The raw link was roughly half as strong as the original finding. Adjusting for family background and early cognitive ability shrank it by about two thirds, leaving most estimates statistically insignificant.

The test is not worthless; it may still reflect something substantive. Research published in Cognition in January 2013 adds context: children whose trust an adult had just betrayed waited markedly less than peers dealing with a reliable adult. How long a child perseveres depends partly on how trustworthy the adults around them seem.

中文译文

想象一下20世纪60年代末至70年代初，斯坦福大学校园里的一所幼儿园。一个四五岁的孩子独自坐在一份零食前，面对一个直截了当的选择：现在吃掉它，或者抵住诱惑，等上约15分钟，就能得到两份。冲动的孩子往往马上放弃等待，意志坚定的孩子则坚持下去。

后续追踪研究让这项测验广为人知。1990年发表于《发展心理学》的一项研究发现，进入青少年时期后，当初等得更久的孩子，往往在美国大学入学考试SAT中取得更高分数。一份不起眼的零食成了美德的象征，四岁时表现出的克制，也被过分推崇为预测人生走向的指标。

然而，这一结论的基础并不牢靠。进入SAT成绩分析的孩子只有几十名，而且主要是斯坦福大学教师和研究生的子女。这样一个背景相似、条件优越的群体，很难代表普通家庭。富裕的家庭环境，也可能同时有助于培养耐心和取得学业成功。

2018年5月发表于《心理科学》的一项研究进行了概念性重复实验：沿用同一研究思路，换用新的样本，并调整研究方法。研究者对来自不同背景的900多名儿童，在四岁半和15岁时分别进行了评估。未经调整的关联强度，大约只有早期研究结果的一半；进一步控制家庭背景和早期认知能力的影响后，这种关联又缩小了约三分之二，大多数估计结果都未达到统计显著性。

这项测验并非毫无价值，它或许仍能反映一些有实质意义的信息。2013年1月发表于《认知》的一项研究补充了背景：刚刚遭遇成年人失信的孩子，等待时间明显短于与守信成年人相处的同龄人。孩子能坚持多久，部分取决于他们觉得身边的大人有多可信。

“统计上不显著”不等于已经证明不存在关联；这里强调，计入背景差异后，早期结论的支持力度明显减弱。

核心词汇

先记这22项。连着下方的原文短语一起回想。

resist v.
抵挡；忍住
resist temptation

temptation n.
诱惑
resist temptation for about 15 minutes

impulsive adj.
冲动的
Impulsive children often surrender

surrender v.
放弃抵抗；屈服
often surrender instantly;

resolute adj.
意志坚定的
resolute ones hold out.

hold out phr. v.
坚持下去
resolute ones hold out.

legendary adj.
广为传颂的；著名的
Follow-ups made the test legendary.

restraint n.
克制；自制
restraint at four was glorified

glorify v.
过分美化；推崇
glorified as a predictor of destiny

predictor n.
预测因素；预测指标
a predictor of destiny.

precarious adj.
不稳固的；不可靠的
Yet the foundation was precarious.

homogeneous adj.
同质的；背景相似的
a homogeneous, privileged group

privileged adj.
享有优越条件的
a homogeneous, privileged group

foster v.
培养；促进
foster patience and academic success

replication n.
重复研究；重复实验
ran a conceptual replication:

diverse adj.
多种多样的
from diverse backgrounds

assess v.
评估
were assessed at four and a half

adjust for phr. v.
控制……的影响
Adjusting for family background

cognitive adj.
认知的
early cognitive ability

insignificant adj.
未达统计显著性的
statistically insignificant.

betray v.
辜负；背弃
trust an adult had just betrayed

trustworthy adj.
可信赖的
how trustworthy the adults around them

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

nursery school n.
幼儿园

campus n.
大学校园

in solitude phr.
独自；独处地

blunt adj.
直截了当的

bargain n.
交易；约定的条件

earn v.
挣得；赢得

instantly adv.
立即；马上

follow-up n.
后续追踪研究

adolescent n.
青少年

attain v.
达到；取得

trivial adj.
微不足道的

snack n.
零食

emblem n.
象征；标志

virtue n.
美德

destiny n.
命运；人生走向

foundation n.
（结论的）基础

a few dozen phr.
几十个

predominantly adv.
主要地；大多

faculty n.
大学全体教员

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

postgraduate adj.
研究生的

scarcely adv.
几乎不

ordinary adj.
普通的

prosperous adj.
富裕的

alike adv.
同样地；两者都

conceptual adj.
概念层面的

modified adj.
调整过的；改动过的

youngster n.
儿童；少年

link n.
关联；联系

statistically adv.
在统计学上

worthless adj.
毫无价值的

reflect v.
反映

substantive adj.
有实质意义的

context n.
背景；具体情境

markedly adv.
明显地

peer n.
同龄人

persevere v.
坚持；不放弃

depend v.
取决于（常与on连用）

on prep.
与depend连用，表示取决于

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Impulsive children often surrender instantly; _____ ones hold out.

02 Yet the foundation was _____.

03 Adjusting for family background and early cognitive ability shrank it by about two thirds, leaving most estimates statistically _____.

04 How long a child perseveres depends partly on how _____ the adults around them seem.

文章主线

- 选择 四五岁孩子可立刻吃一份零食，或等约15分钟换两份，用来观察延迟满足。
- 走红 早期追踪发现，等得久的孩子后来SAT（美国大学入学考试）成绩较高，测验遂被当作成功预测器。
- 局限 早期SAT分析仅数十名孩子，且多来自优裕家庭；家庭条件可能同时影响耐心与成绩。
- 复核 2018年复核900多名不同背景儿童；计入家庭背景与早期认知差异后，关联大幅减弱，多数估计不具统计显著性。
- 关键 另一实验发现，刚遇到不守信大人的孩子等得更短；等待也反映信任，不能据此断定一个人的命运。

参考答案　1 resolute 2 precarious 3 insignificant 4 trustworthy

两度摘得诺奖的女人

The Woman Who Won Two Nobels



居里夫人的两座诺奖从何而来?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Marie Curie was born Maria Skłodowska on 7 November 1867 in Warsaw, then under Russian dominion, where women were denied university admission. Eager to pursue formal study, she emigrated to Paris in 1891, enrolling at the Sorbonne. On 26 July 1895 she wed the physicist Pierre Curie, forging a research partnership.

Henri Becquerel found that uranium emitted rays; she named this property of certain elements radioactivity. In July 1898 the Curies announced polonium, a tribute to her motherland, and on 26 December, radium. In a leaky shed they painstakingly processed tonnes of pitchblende, a uranium ore; by 1902 she had recovered about a tenth of a gram of radium chloride.

On 10 December 1903 she jointly won the physics Nobel with Pierre and Becquerel, the first woman so recognised. The nomination had initially left her out; at Pierre's insistence, the omission was rectified. On 19 April 1906 a horse-drawn wagon fatally struck Pierre. On 5 November she gave her inaugural lecture as his successor, the Sorbonne's first woman professor.

In November 1911 a press scandal about her private life erupted; she was urged to stay away. Undeterred, she collected the chemistry Nobel, won outright, on 10 December. Her distinction in two sciences remains unmatched. In the First World War she mobilised X-ray vans and stationary posts that examined over a million casualties.

She died on 4 July 1934 of a blood disorder, probably caused by prolonged exposure to radiation. Her notebooks, still radioactive, are preserved in lead-lined boxes at the National Library of France; readers sign a liability waiver and don protective garments. The pages retain her handwriting and her radium.

中文译文

玛丽·居里原名玛丽亚·斯克沃多夫斯卡，1867年11月7日生于当时受俄国统治的华沙。那里的大学不招收女性。为了接受正规的高等教育，她于1891年移居巴黎，进入索邦大学。1895年7月26日，她与物理学家皮埃尔·居里结婚，两人也结成了科研搭档。

亨利·贝克勒尔发现铀会发出射线；玛丽把某些元素的这种性质命名为“放射性”。1898年7月，居里夫妇宣布发现钋，以此向她的祖国致敬；同年12月26日，他们又宣布发现镭。两人在一间漏雨的棚屋里，费尽心力处理了成吨的沥青铀矿。到1902年，她提取出了约十分之一克氯化镭。

1903年12月10日，她与皮埃尔、贝克勒尔共同获得诺贝尔物理学奖，成为首位女性诺奖得主。最初的提名没有她的名字，后来在皮埃尔的坚持下，这一遗漏才得到纠正。1906年4月19日，皮埃尔被一辆马拉货车撞倒身亡。同年11月5日，她接替丈夫的职位，发表就职演讲，成为索邦大学首位女教授。

1911年11月，媒体围绕她的私生活掀起一场风波，有人劝她不要前去领奖。她没有退缩，仍于12月10日领取了自己独自获得的诺贝尔化学奖。她在两门科学领域获得诺奖的成就，至今无人比肩。第一次世界大战期间，她组织调用X光车和固定检查站，为超过一百万名伤员做了检查。

1934年7月4日，她因一种血液疾病去世，很可能与长期接触辐射有关。她的笔记本至今仍有放射性，保存在法国国家图书馆内衬铅的盒子里；读者须签署免责协议，穿上防护服才能查阅。书页上既留着她的笔迹，也留着她的镭。

核心词汇

先记这22项。连着下方的原文短语一起回想。

dominion n.
统治；管辖
then under Russian dominion,

deny v.
被拒绝给予；被剥夺
were denied university admission.

admission n.
入学资格；准入
were denied university admission.

pursue v.
追求；从事
Eager to pursue formal study,

emigrate v.
移居国外
she emigrated to Paris in 1891,

enrol v.
注册入学
emigrated to Paris in 1891, enrolling

forge v.
建立；缔结
forging a research partnership.

partnership n.
合作关系
forging a research partnership.

property n.
性质；特性
she named this property of certain

radioactivity n.
放射性
of certain elements radioactivity.

painstakingly adv.
煞费苦心地；精细认真地
they painstakingly processed tonnes of

recover v.
提取；回收
by 1902 she had recovered about a

jointly adv.
共同地
10 December 1903 she jointly won the

insistence n.
坚持；坚决要求
Pierre's insistence, the omission was

omission n.
遗漏；疏漏
Pierre's insistence, the omission was

rectify v.
纠正；改正
the omission was rectified.

inaugural adj.
就职的；首次的
she gave her inaugural lecture as his

successor n.
继任者
inaugural lecture as his successor,

undeterred adj.
未被吓退的
Undeterred,

distinction n.
殊荣；卓越成就
Her distinction in two sciences

mobilise v.
组织调用；动员
World War she mobilised X-ray vans and

prolonged adj.
持续很久的
probably caused by prolonged exposure

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

eager adj.
渴望的；热切的

formal adj.
正规的；正式的

wed v.
与……结婚

ray n.
射线

element n.
化学元素

tribute to n.phr.
对……的致敬

motherland n.
祖国

radium n.
镭

leaky adj.
漏水的；漏雨的

tonne n.
公吨

ore n.
矿石

gram n.
克

chloride n.
氯化物

recognise v.
认可；表彰

initially adv.
最初

leave out phr.v.
与out连用：遗漏；排除

out adv.
与leave连用：排除在外

wagon n.
货运马车

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

fatally adv.
致命地

scandal n.
丑闻；引发非议的风波

erupt v.
突然爆发

outright adv.
独自地；不与他人分享地

unmatched adj.
无人能及的

van n.
厢式汽车

stationary adj.
固定不动的

casualty n.
伤亡人员；此处指伤员

disorder n.
疾病

radioactive adj.
有放射性的

preserve v.
保存

lead n.
铅

liability waiver n.phr.
免责协议

don v.
穿上

protective adj.
防护的

garment n.
衣服；服装

retain v.
保留；留存

handwriting n.
笔迹

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Eager to _____ formal study,

02 at Pierre's insistence, the omission was _____.

03 _____, she collected the chemistry Nobel, won outright, on 10 December.

04 probably caused by _____ exposure to radiation.

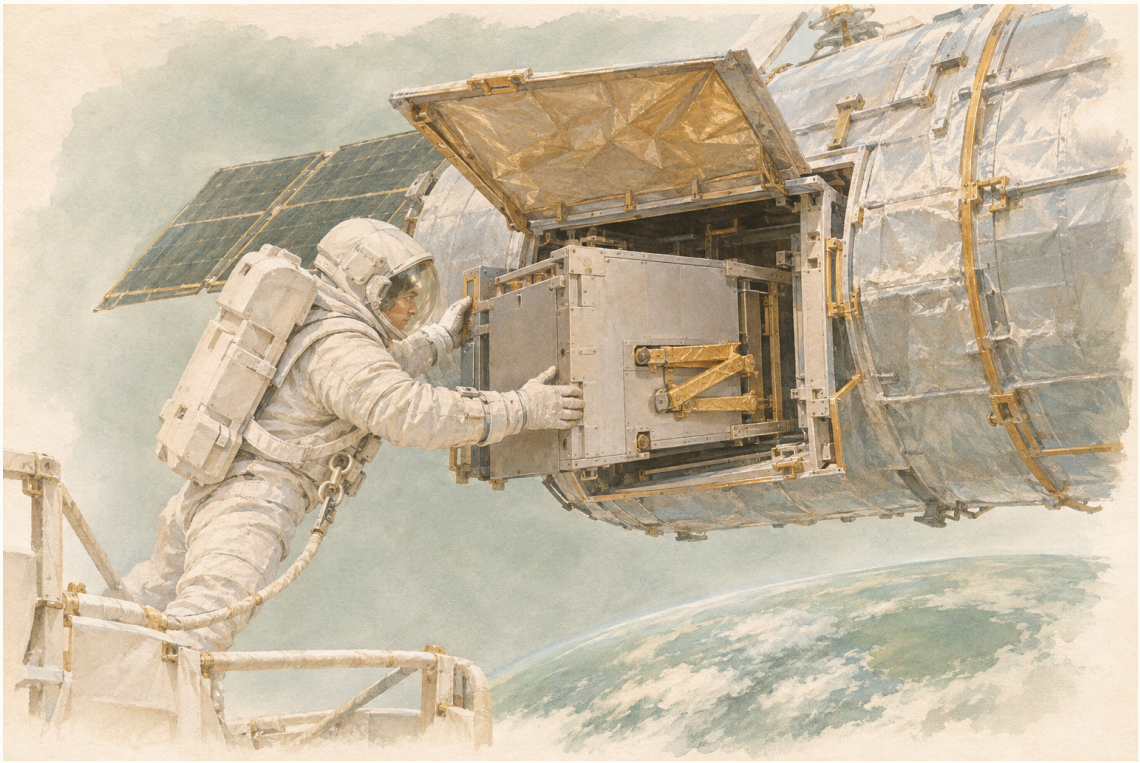
文章主线

- 求学 当时华沙的大学不收女性，玛丽·居里1891年赴巴黎求学，后与皮埃尔结婚合作研究。
- 发现 贝克勒尔发现铀会放出射线，她将这种性质称为放射性；居里夫妇发现钋、镭，从成吨矿石提得约0.1克氯化镭。
- 首奖 1903年，她与皮埃尔、贝克勒尔共享物理学奖，成为首位女性诺奖得主；最初漏提名经皮埃尔坚持纠正。
- 双奖 皮埃尔遇难后，她接任成为索邦首位女教授；1911年顶住私生活风波独获化学奖，仍是唯一获两门科学诺奖者。
- 救护代价 一战时她组织X光设施检查逾百万伤员；后来死于很可能与长期辐射有关的血液病，笔记至今仍有放射性。

参考答案　1 pursue 2 rectified 3 Undeterred 4 prolonged

需要戴眼镜的望远镜

The Telescope That Needed Glasses



主镜没有修好，影像怎样变清晰？

READ 纯净阅读 | CHECK 中文核对 | LEARN 词汇记忆 | RECALL 回想巩固

英文原文

On 24 April 1990 the space shuttle Discovery lofted the Hubble Space Telescope into orbit. The observatory cradled in the payload bay had cost about $1.5 billion, a lavish investment meant to fix its gaze on the cosmos and its galaxies. Money, it soon emerged, could not buy clarity.

Its first snapshots came back blurred and fuzzy. In June 1990 NASA announced a flaw in the main mirror: its outer rim had been flattened by 2.2 micrometres, roughly a fiftieth of the width of a hair. That deviation marred the optics, leaving light without a sharp focus.

The trouble began with calibration. A lens in the assembly that checked the mirror's shape had been offset by about 1.3 millimetres, so the mirror was ground to the wrong curve. A second, independent test by the contractor, Perkin-Elmer, gave a divergent result, but engineers and NASA managers judged it less reliable and left the contradictory reading unexplored.

The mirror could not be repaired in orbit, so engineers prescribed tailored glasses. COSTAR, a corrective cabinet the size of a refrigerator, carried small mirrors on adjustable arms that intercepted the light and compensated for the imperfect mirror's distortion before the light reached the instruments. The mirror itself would stay untouched, and a new main camera was prepared.

On 2 December 1993 Endeavour ascended with seven astronauts. After a rendezvous, they worked in tandem across five spacewalks, about 35 and a half hours of exertion, installing COSTAR and the new camera. The imagery turned crisp, Hubble's vision renewed. The mirror still carries that 2.2-micrometre flaw, and the light it gathers is corrected only after leaving it.

中文译文

1990年4月24日，发现号航天飞机将哈勃空间望远镜送入轨道。这座安放在载荷舱中的太空天文台耗资约15亿美元，是一笔用于凝视宇宙与星系的巨额投资。但人们很快发现，花钱并不一定能买来清晰的影像。

哈勃传回的首批照片模糊不清。1990年6月，美国航天局宣布主镜有缺陷：外缘磨得过平，与正确形状相差2.2微米，约为头发丝宽度的五十分之一。这点偏差损害了光学系统，使光线无法准确聚焦。

问题出在校准上。检测主镜形状的装置中，一块透镜错位了约1.3毫米，主镜因此被磨成了错误的曲面。承包商珀金-埃尔默公司另一项独立测试曾给出不同结果，但工程师和美国航天局管理人员认为该测试较不可靠，没有深入追查那组相互矛盾的读数。

主镜无法在轨道上修复，于是工程师为它设计了一副“定制眼镜”。矫正装置COSTAR的箱体约有冰箱大小，可调机械臂上装着小反射镜，在光线进入仪器前将其截住，抵消有缺陷的主镜造成的畸变。主镜本身保持不动，同时还准备了一台新的主相机。

1993年12月2日，奋进号载着七名宇航员升空。完成交会后，他们协同作业，通过五次太空行走，用约35个半小时安装了COSTAR和新相机。哈勃的影像恢复清晰，仿佛重获视力。但主镜那2.2微米的缺陷仍然存在，光线是在离开主镜之后才得到矫正的。

2.2微米是主镜形状误差，1.3毫米是检测装置透镜的错位；安装矫正装置并未修磨主镜。

核心词汇

先记这22项。连着下方的原文短语一起回想。

observatory n.
天文台
The observatory

payload n.
有效载荷
in the payload bay

lavish adj.
耗资巨大的
a lavish investment

clarity n.
清晰度
could not buy clarity

flaw n.
缺陷
a flaw in the main mirror

deviation n.
偏差
That deviation marred the optics

optics n.
光学系统
marred the optics

calibration n.
校准
The trouble began with calibration

assembly n.
装置；组件
A lens in the assembly

offset v.
使偏离位置
offset by about 1.3 millimetres

contractor n.
承包商
the contractor, Perkin-Elmer

divergent adj.
不同的；不一致的
a divergent result

reliable adj.
可靠的
judged it less reliable

contradictory adj.
相互矛盾的
the contradictory reading

corrective adj.
矫正的
a corrective cabinet

intercept v.
截住
intercepted the light

compensate for 动词短语
弥补；抵消
compensated for

distortion n.
畸变
the imperfect mirror's distortion

untouched adj.
未改动的
The mirror itself would stay untouched

rendezvous n.
交会
After a rendezvous

in tandem 介词短语
协同地
worked in tandem

imagery n.
影像
The imagery turned crisp

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

loft v.
向高处送出

telescope n.
望远镜

into prep.
进入；到……中

cradle v.
稳妥安放

investment n.
投资

gaze n.
凝视；目光

cosmos n.
宇宙

galaxy n.
星系

snapshot n.
照片；快照

blurred adj.
模糊的

fuzzy adj.
模糊不清的

outer adj.
外侧的

rim n.
边缘

flatten v.
磨平；使变平

width n.
宽度

mar v.
损害

sharp adj.
清晰锐利的

lens n.
透镜

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

millimetre n.
毫米

be ground to 动词短语
被磨成

curve n.
曲面；曲线

independent test n.
独立测试

unexplored adj.
未被深入调查的

prescribe v.
开出；提出（方案）

tailored adj.
量身定制的

cabinet n.
箱体

refrigerator n.
冰箱

adjustable adj.
可调节的

imperfect adj.
有缺陷的

ascend v.
升空；上升

spacewalk n.
太空行走

exertion n.
费力；辛苦作业

install v.
安装

crisp adj.
清晰鲜明的

vision n.
视力

renew v.
使恢复

gather v.
收集

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The trouble began with _____.

02 engineers and NASA managers judged it less _____ and left the contradictory reading unexplored.

03 The mirror itself would stay _____, and a new main camera was prepared.

04 The _____ turned crisp, Hubble's vision renewed.

文章主线

- 失焦 1990年，耗资约15亿美元的哈勃望远镜升空，照片却模糊；主镜外缘磨错2.2微米，无法准确聚光。
- 根源 主镜被磨错，是因检测装置的透镜错位约1.3毫米；另一独立测试读数不同，却被认为较不可靠而未追查。
- 方案 主镜无法在轨修复，工程师设计矫正装置COSTAR，用小反射镜在光线进入仪器前抵消畸变，并另备新相机。
- 修复 1993年，七名宇航员在五次太空行走中协同作业约35.5小时，装上COSTAR和新相机。
- 结果 影像恢复清晰，但主镜本身始终未动，2.2微米缺陷依旧；被矫正的是离开主镜后的光线。

参考答案　1 calibration 2 reliable 3 untouched 4 imagery